User Study

Perception of Virtual Avatar Advisors

Scene-by-Scene Study Flow

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Mori (1970) — The Uncanny Valley

Mori (1970)

Foundational theoretical paper — proposes the human affinity curve for robots and artificial figures

Key Findings

The valley: affinity rises with human-likeness, then drops sharply at the near-human threshold, recovering only at complete fidelity

The problem zone: modern production avatars sit at mid-range realism — exactly where the drop is deepest and discomfort is strongest

Movement amplifies: a moving near-human triggers stronger aversion than a static one at the same realism level — critical for animated avatars

What this study adopts

Core motivation for NPR: stylisation deliberately places the avatar below the uncanny zone, trading photorealistic fidelity for affective safety. Directly motivates Research Questions 1 and 3.

Canales, Roble & Neff (2024) — Avatar Stylisation and Trust

Canales et al. (2024)

Virtual doctor avatars at three stylisation levels; video-based trust study with forced-choice advisor selection

Key Findings

Three conditions: realistic, semi-realistic, caricatured virtual doctor — same script, same camera, same voice

Self-report trust: statistically equivalent across all three stylisation levels — no significant differences between conditions

Behavioural selection: semi-realistic attracted the highest rate of "which doctor would you consult?" choices

Dissociation: explicit trust ratings did not predict implicit advisor choice — stylisation affects implicit social judgment

What this study adopts

Entire study paradigm: video-based, scripted advisor, same voice and camera. Trust battery structured around McKnight dimensions. Post-study advisor preference ranking (1st/2nd/3rd) as the behavioural measure — adapted from forced-choice to preserve C2 vs C3 signal under expected C1 dominance.

McKnight & Chervany (2001) — What Trust Means

McKnight & Chervany (2001)

Proposes that trust is not one score but three independent, measurable dimensions

Key Framework

Insight: trust has three independent dimensions (competence, benevolence, integrity), not a single score — rendering may affect each differently

Competence: perceived ability to deliver advice ("does this advisor actually know careers and routes?")

Benevolence: perceived intent to act in YOUR interest ("is this advisor giving advice that helps me?")

Integrity: perceived adherence to honest and principled norms ("is this advisor straight with me, even unwelcome advice?")

What this study adopts

11-item trust battery: 4 competence items + 4 benevolence items + 3 integrity items (Parts E, F, G). Each subscale scored separately. Cronbach's α computed per condition.

Bartneck et al. (2009) — Godspeed Questionnaire

Bartneck et al. (2009)

Five validated semantic-differential subscales for human-robot and avatar interaction research

Godspeed Subscales

Five subscales (5 items, 5-point each): Anthropomorphism, Animacy, Likeability, Perceived Intelligence, Perceived Safety

Likeability (Part B): affective warmth and appeal — proxy for social acceptance and the affective component of trust

Anthropomorphism (Part C): perceived human-likeness — does NPR reduce eeriness while preserving human-likeness?

We do NOT use: animacy, perceived intelligence, or perceived safety — outside this study's scope, adds measurement fatigue

What this study adopts

Part B — 5 Likeability items (5-point). Part C — 5 Anthropomorphism items (5-point). Anthropomorphism tests whether NPR mediates eeriness by changing perceived humanness.

Ho & MacDorman (2010) — Eeriness Scale

Ho & MacDorman (2010)

Validated alternative to Godspeed for uncanny valley research; separates eeriness from human-likeness

Key Findings

The Godspeed problem: existing eeriness items conflated discomfort with human-likeness — two dimensions that can vary independently

Key distinction: an avatar can look human without feeling eerie, or feel eerie while appearing non-human — separate perceptual dimensions

The instrument: 6 semantic-differential items, 5-point, two subscales (eeriness and human-likeness) validated across robotic and virtual stimuli

What this study adopts

Part D — 6 eeriness items, 5-point (adapted). Directly tests H1: does NPR reduce eeriness? Eeriness and human-likeness are measured separately to detect if NPR reduces uncanniness without reducing perceived human-likeness.

Alipour et al. (2025) — Systematic Review of 53 Studies

Alipour et al. (2025)

53 empirical studies at the intersection of uncanny valley effects and trust in human-agent interaction

Critical Findings

Flaw 1 — static images: most studies used static images, not video — invalid for animated avatars whose eeriness is motion-dependent

Flaw 2 — single score: most collapsed trust to one number, masking differential effects across competence, benevolence, and integrity

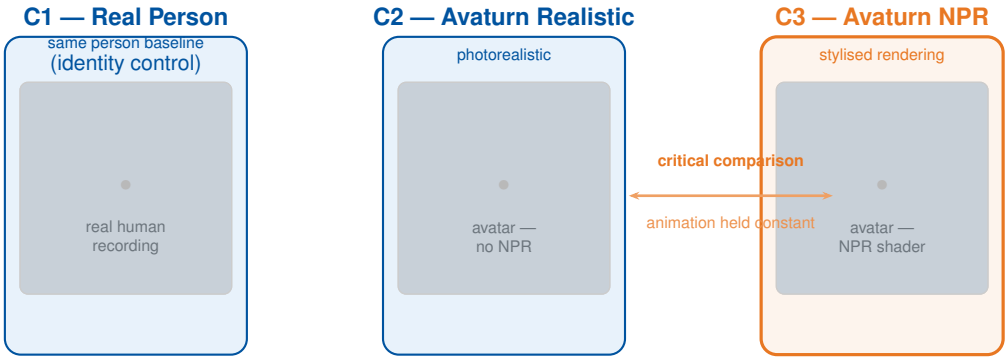
Flaw 3 — no behavioural measure: self-report alone is insufficient; Canales (2024) showed explicit ratings and implicit choice can diverge

Prescription: valid studies need dynamic video stimuli, multi-component trust battery, and at least one behavioural indicator

What this study adopts

This review directly justifies every design decision: video-based stimuli (not static), 3-subscale McKnight trust battery, and post-study advisor selection as the behavioural indicator.

Three Study Conditions



same person • same voice • same script • only rendering varies (C2 vs C3)

Three Advisor Scenarios

Scenario	Context	Option A	Option B	Recommends
S1 Career Choice — Competence	Master's degree just completed. Two job offers, one month start. No financial pressure. Five days to decide.	**Job A** Generalist role, small company. Broad scope, high individual visibility. Skills transfer across industries. **Salary approx. 8 000 EUR/yr less.**	**Job B** Specialist role, large well-known company. Higher salary, recognised brand. Deep specialisation within a large team, less individual visibility.	**Job A** *"Pivot option won't come back later"*
S2 Travel Route — Benevolence Competence	Arrived in unfamiliar city. Must reach hotel in one hour for a colleague meeting. Two public transport routes found.	**Route A** 50 min, direct line, no transfers. Arrives 10 min before meeting. **Reliable.**	**Route B** 20 min, one transfer at an unfamiliar station. Arrives 40 min early *if the connection holds.*	**Route B** *"Buffer beats a tight margin"*
S3 Learning Resource — Integrity	Wants to learn data analysis for CV. A few weeks available, two options found. Must commit to one.	**Option A** Structured course, 35 EUR, 8 hrs. **CV certificate from named platform.**	**Option B** Free tutorial series by a practitioner. Same content, flexible pace. No certificate.	**Option A** *"Certificate signals the skill on a CV"*

What the Avatar Says — Recommendation Scripts

S1
Career Choice
→ Job A

"I'd go with Job A. Yes, the salary is lower right now, but the generalist role gives you a range of experience that is genuinely hard to get later. Salary gaps can be closed. But once you specialise, switching back is costly. Take the pivot while you still can."

S2
Travel Route
→ Route B

"I'd take Route B. Yes, it has a transfer at a station you have never used. But it gets you there forty minutes early. If something goes wrong, you still have time to recover. Route A is direct but leaves only ten minutes. That is not enough. Take the buffer."

S3
Learning Resource
→ Option A

"I would go with the paid course. Yes, 350 euros is a real cost. But the certificate is what matters here. It is not just about what you learn, it is about what you can show. A recruiter cannot verify a free tutorial series. That is worth it."

Video Production Checklist (researcher note — before recording)

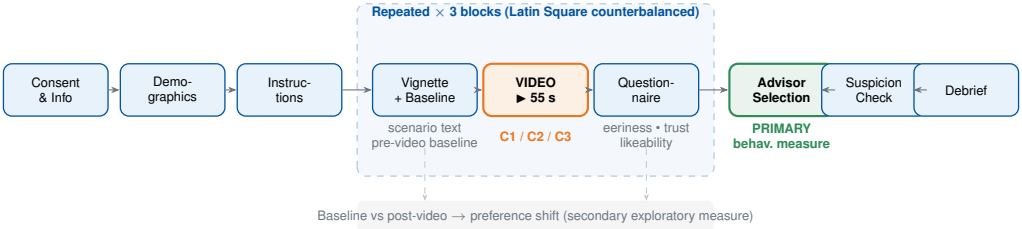
Match across C1, C2, C3

- **Resolution & ratio:** 1920×1080, 16:9 for all three
- **Frame rate:** 30 fps — Unity export must match camera
- **Duration:** within ±1 s per scenario
- **Framing:** eye-level, centred, same head-to-shoulder crop
- **Background:** same desk, setting, camera angle replicated in Unity scene

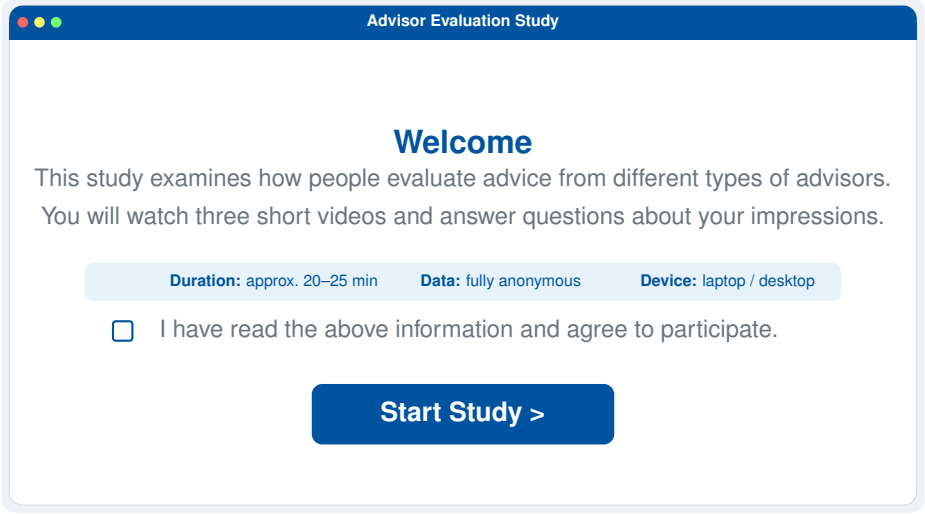
Audio & technical

- **Audio:** one recording, same mic and room, applied to all three; normalise to −16 LUFS
- **No TTS** in any condition
- **Lighting colour:** match Unity light temperature to C1 camera environment
- **Codec:** H.264 MP4, same bitrate for all three

Study Flow Overview



Scene 1 — Welcome & Consent



Scene 2 — Participant Instruction

Advisor Evaluation Study

Please complete this study on a **laptop or desktop computer**. A phone screen is too small for accurate viewing. **If you are currently on a mobile phone or tablet, please switch devices before continuing.**

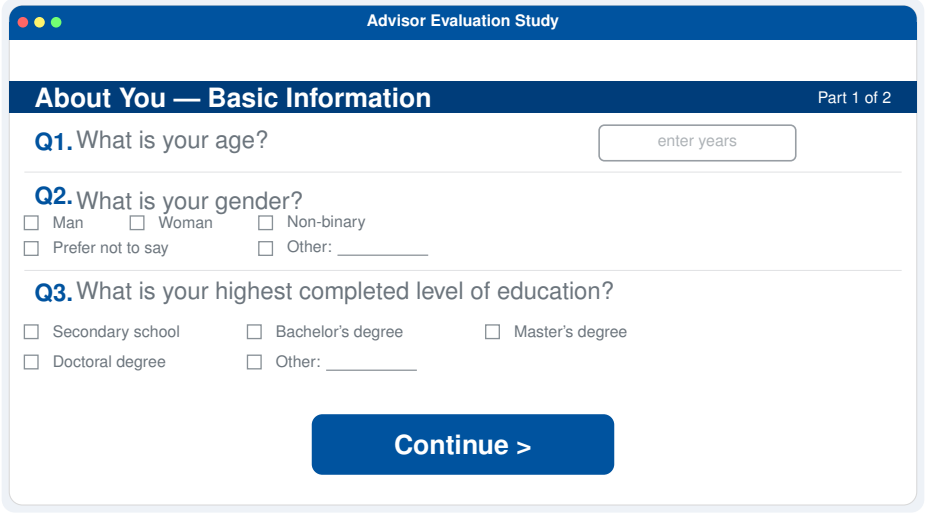
You will watch **three short videos**. Each video shows a virtual advisor making a recommendation. After each video, you will answer a set of questions about your immediate impression of the advisor.

There are no right or wrong answers. Please respond based on your honest first impression, not what you think the "correct" answer is.

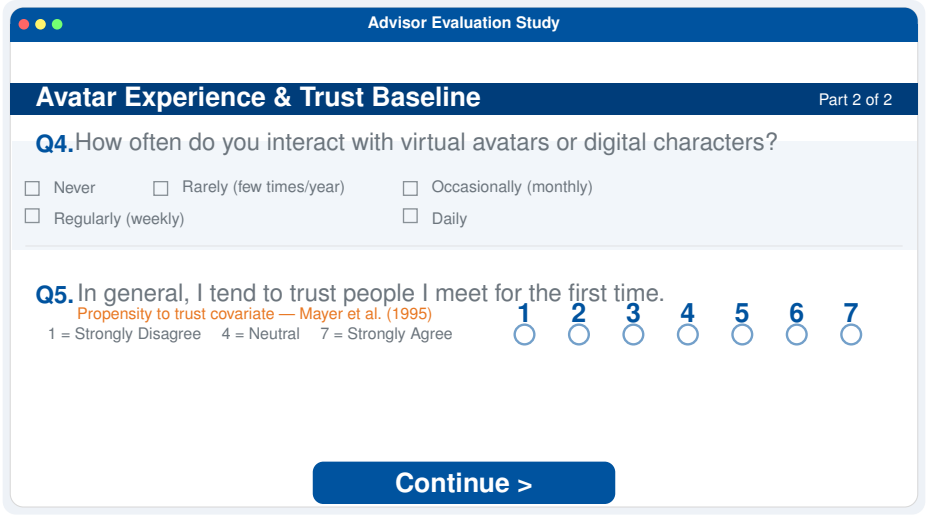
Please watch each video in full before answering.

Continue >

Scene 2b — Demographics (1 of 2)



Scene 2c — Demographics (2 of 2)



Scene 3 — Scenario Page (Career Choice)

Scenario 1 of 3 — Before You Watch

Scenario Context (read before watching)

You recently completed your master's degree and have received two job offers in your field. Both positions start in one month. You have no strong financial pressure and both cities are equally acceptable to you. You have five days to decide.

Job A

A broad generalist role at a smaller company. The work area is yours to define, the team is small, and your contributions are directly visible. Skills transfer across industries. **Starting salary approx. 8 000 EUR/yr less than Job B.**

Job B

A specialist role at a large, well-known company in a high-demand field. The salary is substantially higher and the employer name carries weight on a CV. The role involves deep specialisation within a large team, with less individual visibility.

Before watching — which option would you initially lean toward?

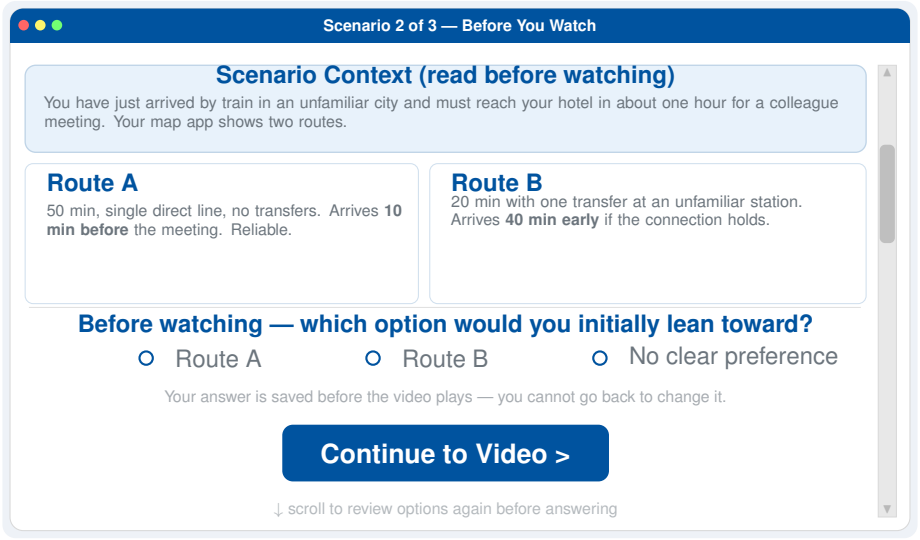
○ Job A ○ Job B ○ No clear preference

Your answer is saved before the video plays — you cannot go back to change it.

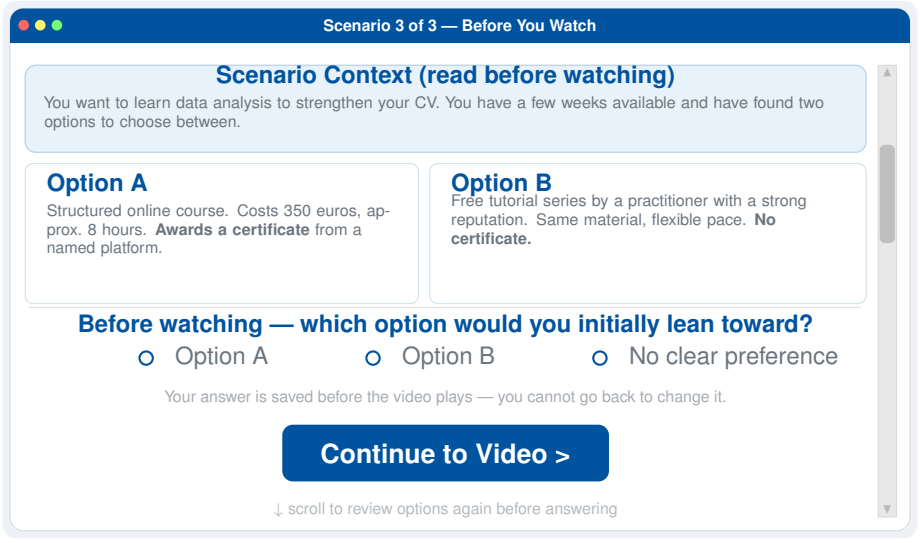
Continue to Video >

↓ scroll to review options again before answering

Scene 3b — Scenario Page (Travel Route)



Scene 3c — Scenario Page (Learning Resource)



Scene 4 — Block Structure & Part A Comprehension Check

This block repeats × 3. Videos are always labelled "Video 1 / 2 / 3" — condition names C1/C2/C3 are never shown.



= 31 closed items per block

Part A. Video Comprehension Check (exclusion criterion: wrong answer or "did not watch" → exclude)

A1. What did the advisor in this video recommend?

○ Option A ○ Option B ○ I did not watch the full video

Answer options configured per scenario in SoSci Survey.

Scene 5 — Parts B & C: Likeability + Human-Likeness (Godspeed)

Part B. Likeability — 5 items

		1	2	3	4	5	
B1	*Dislike*	○	○	○	○	○	*Like*
B2	*Unfriendly*	○	○	○	○	○	*Friendly*
B3	*Unkind*	○	○	○	○	○	*Kind*
B4	*Unpleasant*	○	○	○	○	○	*Pleasant*
B5	*Awful*	○	○	○	○	○	*Nice*

B score: mean(B1–B5). All 5 items verbatim from Godspeed III (Bartneck et al., 2009). No modifications.

Part C. Human-Likeness — 5 items

		1	2	3	4	5	
C1	*Fake*	○	○	○	○	○	*Natural*
C2	*Machinelike*	○	○	○	○	○	*Humanlike*
C3	*Unconscious*	○	○	○	○	○	*Conscious*
C4	*Artificial*	○	○	○	○	○	*Lifelike*
C5	*Moving rigidly*	○	○	○	○	○	*Moving elegantly*

C score: mean(C1–C5). All 5 items verbatim from Godspeed I (Bartneck et al., 2009). C5 anchor is *Moving elegantly* (not "naturally") — exact original wording.

Scene 6 — Part D: Eeriness (Ho & MacDorman, 2010 — 6 items, 5-point)

Rate your impression of the advisor	1 = negative anchor	2	3	4	5 = positive anchor
	1	2	3	4	5
D1	*Not at all eerie*	○	○	○	○	○	*Very eerie*
D2	*Not at all creepy*	○	○	○	○	○	*Very creepy*
D3	*Predictable*	○	○	○	○	○	*Unsettling*
D4	*Ordinary*	○	○	○	○	○	*Supernatural*
D5	*Comfortable to watch*	○	○	○	○	○	*Disturbing to watch*
D6	*Familiar*	○	○	○	○	○	*Feels out of place*

Scene 7 — Part E: Trust Competence (Canales et al., 2024 — 4 items, 7-point Likert)

Part E. Trust: Competence — 4 items (7-point Likert, 1 = SD, 7 = SA)

1 = Strongly Disagree 4 = Neutral 7 = Strongly Agree

	1	2	3	4	5	6	7
E1. I believe this advisor is capable and competent.	○	○	○	○	○	○	○
E2. I would rely on advice given by this advisor.	○	○	○	○	○	○	○
E3. This advisor seems knowledgeable.	○	○	○	○	○	○	○
E4. This advisor seems to have the expertise needed to give sound advice on this kind of decision.	○	○	○	○	○	○	○

Source: Canales et al. (2024); McKnight & Chervany (2001)

E score: mean(E1–E4) Higher = stronger perceived competence

Scale: 1 = Strongly Disagree 4 = Neither agree nor disagree 7 = Strongly Agree

Tests H2: Does NPR rendering (C3) produce significantly lower competence ratings than photorealistic avatar (C2)? Equivalence is predicted — self-reported competence trust should not differ by rendering style.

Scene 8 — Part F: Trust Benevolence (Canales et al., 2024 — 4 items, 7-point Likert)

Part F. Trust: Benevolence — 4 items (7-point Likert, 1 = SD, 7 = SA)

1 = Strongly Disagree 4 = Neutral 7 = Strongly Agree

	1	2	3	4	5	6	7
F1. I feel comfortable in the presence of this advisor.	○	○	○	○	○	○	○
F2. This advisor appears to have good intentions.	○	○	○	○	○	○	○
F3. I believe this advisor would act in my best interest.	○	○	○	○	○	○	○
F4. I felt this advisor was looking out for my best interest, not just giving a generic answer.	○	○	○	○	○	○	○

Source: Canales et al. (2024); McKnight & Chervany (2001)

F score: mean(F1–F4) Benevolence = warmth, caring, acting in the participant's interest

Scale: 1 = Strongly Disagree 4 = Neither agree nor disagree 7 = Strongly Agree

Note: Benevolence may be most sensitive to eeriness via affective social inference.

Scene 9 — Part G: Trust Integrity (Canales et al., 2024 — 3 items, 7-point Likert)

Part G. Trust: Integrity — 3 items (7-point Likert, 1 = SD, 7 = SA)

1 = Strongly Disagree 4 = Neutral 7 = Strongly Agree

	1	2	3	4	5	6	7
G1. This advisor reasons transparently, acknowledging trade-offs rather than oversimplifying.	○	○	○	○	○	○	○
G2. This advisor appears to be honest.	○	○	○	○	○	○	○
G3. I believe this advisor would keep its word.	○	○	○	○	○	○	○

Source: Canales et al. (2024); McKnight & Chervany (2001)

G score: mean(G1–G3) Higher = stronger perceived integrity

Scale: 1 = Strongly Disagree 4 = Neither agree nor disagree 7 = Strongly Agree

Integrity captures transparency and intellectual honesty. G1 directly tests whether the advisor acknowledges trade-offs rather than oversimplifying recommendations.

Scene 10 — Part H: Recommendation Response + Attention Check

Part H. Response to the Recommendation — 1 item (7-point Likert)

1 = Strongly Disagree 4 = Neutral 7 = Strongly Agree

1 2 3 4 5 6 7

H1. How much do you agree with the recommendation this advisor gave?

○ ○ ○ ○ ○ ○ ○

H1 measures recommendation congruence: did the advisor's reasoning shift the participant's preference? Used for the within-block preference-shift analysis (compare pre-video baseline to H1). Not compared across conditions directly (scenario content confound). H2 (warmth) was removed as redundant with Part B Likeability.

AC. Attention Check — embedded once per block

"To confirm you are reading carefully, please select option **[N]** regardless of anything else."

Target value varies per block:

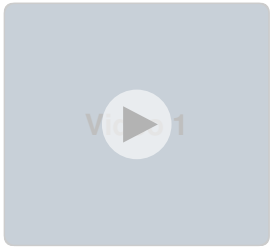
Block 1: select 3 Block 2: select 4 Block 3: select 2

Participants who fail any one check are excluded from all analyses.

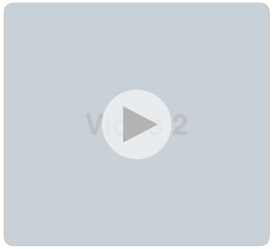
Scene 11 — Post-Study: Advisor Preference Ranking (P1)

★ P1. Primary Behavioural Measure (adapted from Canales et al., 2024)

Imagine each advisor advising on **any** topic — not only the scenario you saw. Rank from **most** (1) to **least** (3) preferred to consult. Assign each advisor a different rank.



Rank: _____



Rank: _____



Rank: _____

In SoSci Survey: drag-and-drop rank order question. Advisor labels show as "Video 1 / 2 / 3" — condition names (C1/C2/C3) are never revealed to participants. Latin Square group mapping applied during data cleaning.

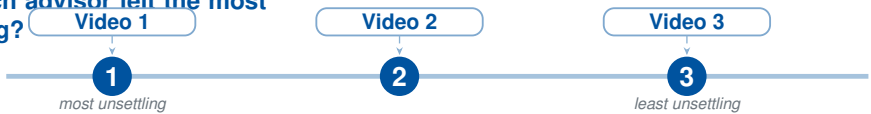
Scene 12 — Comparative Preference Ranking (P2, P3, P4)

Post-Study — drag advisor chips to desired rank position

P2. Which advisor felt the most trustworthy



P3. Which advisor felt the most unsettling?



P4. Which advisor suits a professional context most?



Post-Study — Open Reflection (P5, P6, J)

Post-Study — Open Reflection

P5. Looking back at all three advisors, what was the most important factor in deciding which one you trusted the most?

open text response

P6. Did the visual appearance of any advisor distract you from the content of the recommendation? If yes, describe.

open text response

J. Was there a specific moment, comment, or visual detail in any of the three videos that noticeably increased or decreased your trust in that advisor? If yes, please describe briefly.

open text response

Responses used for exploratory qualitative analysis

P5 and P6 are open-ended and optional. J is post-study and covers all three videos. Responses supplement the quantitative scales. Responses will be analysed thematically and reported alongside quantitative findings. Optional — participants who skip are not excluded.

Post-Study — Part I(a): Visual Style Check — Stylisation Rating (I1)

Part I. Visual Style Check — Manipulation Check

Shown ONCE after all three videos. Per-block placement would prime participants to attend to visual style.

I1. How visually stylised or non-realistic did each advisor appear?

Scale: 1 = Completely realistic 7 = Very stylised

	Completely realistic (1)						7
	1	**2**	**3**	**4**	**5**	**6**	
I1a. Video 1 (Advisor 1)	○	○	○	○	○	○	○
I1b. Video 2 (Advisor 2)	○	○	○	○	○	○	○
I1c. Video 3 (Advisor 3)	○	○	○	○	○	○	○

I1 score: one rating per advisor. Expected pattern: C3 (NPR) scores highest, C1 (real person) lowest. A significant difference confirms the rendering manipulation was perceived as intended.

Post-Study — Part I(b): Visual Style Check — Realism Judgement (I2)

Part I. Visual Style Check — Manipulation Check (continued)

I2. The visual appearance of this advisor was noticeably artificial or non-realistic.

Scale: 1 = Strongly Disagree … 7 = Strongly Agree

	Strongly Disagree (1)						7
	1	2	3	4	5	6	7
I2a. Video 1 (Advisor 1)	○	○	○	○	○	○	○
I2b. Video 2 (Advisor 2)	○	○	○	○	○	○	○
I2c. Video 3 (Advisor 3)	○	○	○	○	○	○	○

I2 score: one rating per advisor. Complements I1 with an agree/disagree format. Together I1 and I2 confirm whether participants perceived the rendering difference between C2 (photorealistic) and C3 (NPR).

Post-Study — Appearance Influence (HA)

HA. Appearance Influence (rate all 3 advisors)

For each advisor, to what extent did their visual appearance influence how you felt about their recommendation?

1 = Not at all 4 = Somewhat 7 = Very strongly

1 = Not at all 4 = Somewhat 7 = Very strongly

	1	2	3	4	5	6	7
HA1. Advisor from Video 1	○	○	○	○	○	○	○
HA2. Advisor from Video 2	○	○	○	○	○	○	○
HA3. Advisor from Video 3	○	○	○	○	○	○	○

HA score: rated per advisor. High HA + lower trust in C3 supports a visual salience explanation. Low HA + lower trust suggests an automatic uncanny valley response not mediated by conscious attention to style.

Scene 6 — Debrief

